

On Erdős's Multiplicative Representation Problem

Rishikesh Gajjala
NYU Abu Dhabi

Abstract

Let $g_3(n)$ be the largest size of a set $A \subseteq [n]$ for which every integer has at most two representations $m = ab$ with $a, b \in A$ and $a < b$. We prove

$$g_3(n) = \frac{n \log \log n}{\log n} + (1 + M + \Gamma + o(1)) \frac{n}{\log n},$$

where M is the Meissel–Mertens constant and Γ is a finite constant defined by a discrete cofactor problem. This resolves a five-decade-old problem posed by Erdős. The initial proofs were generated and formalized in Lean by OpenAI's GPT 5.6 Sol; the authors have verified and rewritten proofs to enhance readability and provide additional context.

2020 Mathematics Subject Classification. Primary 11B83; secondary 05D05, 11N37.

Keywords. Multiplicative representation, multiplicative Sidon set, extremal hypergraph, semiprime, variational constant.

1 Introduction

Write $[n] = \{1, \dots, n\}$, and let

$$\rho_A(m) = \#\{(a, b) : a, b \in A, a < b, ab = m\}$$

and

$$g_3(n) = \max\{|A| : A \subseteq [n], \rho_A(m) \leq 2 \text{ for all } m\}.$$

Erdős proved [4] that

$$g_3(n) \sim \frac{n \log \log n}{\log n}.$$

Erdős posed the second-order question underlying Problem 796 in his 1969 paper *Problems and results in chromatic graph theory* [5]. In its corrected modern form [1], it asks whether

$$g_3(n) = \frac{n \log \log n}{\log n} + (c + o(1)) \frac{n}{\log n}.$$

Let

$$M = \lim_{x \rightarrow \infty} \left(\sum_{p \leq x} \frac{1}{p} - \log \log x \right)$$

be the Meissel–Mertens constant. Our main result is the following.

In a January 2026 note, Tang [9] identified the corrected second-order scale and gave a construction proving

$$g_3(n) \geq \frac{n \log \log n}{\log n} + (1 + M + o(1)) \frac{n}{\log n}.$$

We prove that the second-order constant exists, identify it as $1 + M + \Gamma$, and give the explicit bounds

$$\frac{4}{15} \leq \Gamma < 13.$$

Theorem 1.1. *There is a finite constant Γ satisfying*

$$\frac{4}{15} \leq \Gamma < 13$$

such that

$$g_3(n) = \frac{n \log \log n}{\log n} + (1 + M + \Gamma + o(1)) \frac{n}{\log n}.$$

Thus the answer is yes, with $c = 1 + M + \Gamma$, and

$$1.5281638795 \dots = 1 + M + \frac{4}{15} \leq c < 15.$$

The proof has three parts. First, a pruning argument leaves two normal forms: a large prime times a smooth cofactor, and a small multiplier times two large primes. A C_4 -overlap argument absorbs the second form into the semiprime baseline, while the first is encoded by compatible cofactor families. Finally, the resulting finite optimization problems converge to a fixed constant Γ .

The pruning and complete-box mechanism refine Erdős's proof of the leading-order asymptotic [4, Lemma, pp. 252–253; proof of Theorem 3, pp. 255–261]. Erdős already split integers by the sizes of their factors and used a multipartite hypergraph box to force many equal products. Here that strategy is sharpened to retain terms of order $n/\log n$. The additional ingredients are the C_4 -overlap normalization, the compatible-cofactor variational model, the global collision cleaning, and the finite certificate. The formulation and its correction history are summarized in [1].

The Mertens component follows the Abel-summation route in Goldmakher's *A quick proof of Mertens' theorem* [6], as formalized in the `PrimeNumberTheoremAnd` library. The accompanying Lean development, including the explicit numerical certificates, is available at <https://github.com/rishigajjala/erdos-796-lean>.

Throughout the paper, all logarithms are natural. The functions $\pi(x)$ and $P^+(a)$ denote the prime-counting function and the largest prime factor of a , respectively, with $P^+(1) = 1$. An integer is z -smooth if all its prime factors are at most z ; thus 1 is z -smooth. Constants implicit in $O(\cdot)$ and $\ll$ are absolute unless a dependence is displayed.

2 Compatible cofactor families

For finite sets $U, V \subseteq \mathbb{N}$, define the ordered product convolution

$$r_{U,V}(m) = \#\{(u, v) \in U \times V : uv = m\}.$$

Definition 2.1. A sequence $\mathcal{U} = (U_j)_{j \geq 1}$ is *compatible* if

$$U_j \subseteq [j] \quad \text{and} \quad r_{U_i, U_j}(m) \leq 2 \quad (i, j, m \geq 1).$$

When $i = j$, compatibility says that U_i has at most one unordered factorization of any integer, with diagonal factorizations included. Indeed, two distinct off-diagonal factorizations give four ordered pairs, whereas a diagonal and an off-diagonal factorization give three.

For $j \geq 1$, put

$$N_j(n) = \# \left\{ \sqrt{n} < q \leq n : q \text{ prime and } \left\lfloor \frac{n}{q} \right\rfloor = j \right\}, \quad (1)$$

and define

$$G(n) = \max_{\mathcal{U} \text{ compatible}} \sum_{j \geq 1} N_j(n) |U_j|. \quad (2)$$

Only $j < \sqrt{n}$ have nonzero weight, so (2) is a finite optimization problem.

Lemma 2.2 (The model lower bound). *For every compatible family $\mathcal{U}$, the set*

$$A_n(\mathcal{U}) = \left\{ qu : \sqrt{n} < q \leq n \text{ prime, } u \in U_{\lfloor n/q \rfloor} \right\} \quad (3)$$

is admissible for $g_3(n)$. Consequently $G(n) \leq g_3(n)$.

Proof. If $u \leq n/q < \sqrt{n}$, then q is the unique prime divisor of qu exceeding $\sqrt{n}$. In particular, the parametrization in (3) is injective.

Consider a product of two members of $A_n(\mathcal{U})$. If their large primes are distinct, say q and Q , every representation identifies which factor contains q and which contains Q . It therefore injects into

$$U_{\lfloor n/q \rfloor} \times U_{\lfloor n/Q \rfloor},$$

and compatibility permits at most two ordered cofactor pairs. If both large primes equal q , representations correspond to unordered off-diagonal factorizations inside one U_j , of which there is at most one. All factor pairs counted here consist of distinct integers, as required in the definition of $g_3(n)$. $\square$

The reverse inequality, up to the required error, is the structural heart of the proof.

Theorem 2.3 (Structural reduction). *As $n \rightarrow \infty$,*

$$g_3(n) = G(n) + o\left(\frac{n}{\log n}\right).$$

3 A complete-box estimate

Let $K_{2,2,2}^{(3)}$ denote the complete tripartite 3-uniform hypergraph with two vertices in each part.

Lemma 3.1 (Unbalanced complete-box bound). *If a tripartite 3-uniform hypergraph H , with positive part sizes n_1, n_2, n_3 , contains no $K_{2,2,2}^{(3)}$, then*

$$e(H) \ll \frac{n_1 n_2 n_3}{\min(n_1, n_2, n_3)^{1/4}}.$$

Proof. Write the part sizes as $x \leq y \leq z$, and let $d(u, v)$ be the codegree of $(u, v) \in Y \times Z$ into the smallest part. For each pair of vertices in the smallest part, its common link in $Y \times Z$ is C_4 -free. Summing the Kővári–Sós–Turán bound [7] over those pairs gives

$$S := \sum_{Y \times Z} \binom{d(u, v)}{2} \leq x^2 z \sqrt{y}.$$

Cauchy–Schwarz then gives

$$\frac{e(H)^2}{yz} \leq \sum_{Y \times Z} d(u, v)^2 = e(H) + 2S.$$

Therefore

$$e(H) \leq yz + \sqrt{2} xy^{3/4} z \leq (1 + \sqrt{2}) x^{3/4} yz,$$

as required. $\square$

4 Structural reduction

Put $L = \log n$ and

$$Y = L^{24}, \quad Z = L, \quad W = L^{1/8}, \quad R = Y^4 = L^{96}. \quad (4)$$

For large n ,

$$1 < W < Z < Y < R < \sqrt{n}.$$

We use one simple observation repeatedly. Suppose triples (x, y, z) parametrize distinct elements $xyz \in A$. Their tripartite hypergraph cannot contain $K_{2,2,2}^{(3)}$: the eight elements in such a cube would give the four complementary representations

$$\begin{aligned} (x_0 y_0 z_0)(x_1 y_1 z_1), & \quad (x_0 y_0 z_1)(x_1 y_1 z_0), \\ (x_0 y_1 z_0)(x_1 y_0 z_1), & \quad (x_0 y_1 z_1)(x_1 y_0 z_0) \end{aligned}$$

of the same integer. Injectivity makes these four distinct-factor representations.

Lemma 4.1 (Pruning to two normal forms). *After deleting $o(n/L)$ elements from an admissible $A \subseteq [n]$, every remaining element has exactly one of the forms*

$$\begin{aligned} \text{(S)} \quad & qt, \quad q > \sqrt{n} \text{ prime}, \quad t < R, \quad P^+(t) \leq Z; \\ \text{(P)} \quad & sqr, \quad s < W, \quad q > r > Z \text{ primes}, \quad q > Y. \end{aligned}$$

All estimates are uniform in A .

Proof. First discard elements having a factorization xyz with $x, y, z > Y$. Choose one such factorization for each element and place its coordinates in full dyadic ambient intervals. By the observation above and Lemma 3.1, each box contains $O(nY^{-1/4})$ elements. There are $O(L^3)$ boxes, so the discarded set has size

$$O(nL^3 Y^{-1/4}) = O(n/L^3) = o(n/L).$$

A remaining Y -smooth integer is at most Y^6 . Otherwise we could greedily extract two divisors in $(Y, Y^2]$, leaving a third factor larger than Y . Thus all remaining Y -smooth integers contribute at most

$$Y^6 = L^{144} = o(n/L).$$

For every other element put $q = P^+(a) > Y$ and $t = a/q$. If t is Z -smooth, then $t < R$. Indeed, if $t \geq Y^4$, a minimal subproduct $d > Y$ satisfies $d \leq YZ < Y^2$, and $q, d, t/d$ are three factors larger than Y . The cases $q \leq \sqrt{n}$ contribute at most $R\pi(\sqrt{n}) = o(n/L)$, leaving form (S).

If t is not Z -smooth, put $r = P^+(t) > Z$ and $s = t/r$. Then $q \geq r$ and $s < R$. To see the latter, suppose $s \geq Y^4$. If $r > Y$, the factors q, r, s are all larger than Y . If $r \leq Y$, then s is Y -smooth; a minimal divisor $Y < d \leq Y^2$ makes $q, d, rs/d$ three factors larger than Y . Both cases were already discarded. The case $q = r$ contributes at most $R\pi(\sqrt{n}) = o(n/L)$.

It remains to discard $W \leq s < R$. The triples (s, q, r) are canonical, since $q = P^+(a)$, $r = P^+(a/q)$, and $q > r$; hence they parametrize their products injectively. In a dyadic box of scales S, Q, P , the three full ambient parts have upper orders

$$O(S), \quad O(Q/\log Q), \quad O(P/\log P),$$

and the prime number theorem in dyadic intervals gives the corresponding lower orders for the two prime parts. Since $S \geq W$, $P \geq Z$, and $Q \geq Y$, each part has size $\gg W$. Lemma 3.1 gives

$$e(S, Q, P) \ll \frac{SPQ}{\log P \log Q} W^{-1/4}.$$

In every nonempty box $Q \gg P$, so $P \ll \sqrt{n/S}$ and therefore $\log(n/(SP)) \geq \frac{1}{2} \log(n/S) + O(1) \asymp L$. For fixed S, P , summing the geometric Q -scales gives

$$\sum_Q e(S, Q, P) \ll \frac{n}{L \log P} W^{-1/4}.$$

There are $O(\log \log n)$ dyadic S -scales, and summing $1/\log P$ over the dyadic P -scales contributes another $O(\log \log n)$. Hence

$$\sum_{S, Q, P} e(S, Q, P) \ll \frac{n}{L} W^{-1/4} (\log \log n)^2 = o(n/L).$$

The remaining elements are exactly those of forms (S) and (P). □

4.1 The semiprime form

Let A_{sp} be the elements of form (P), and for $s < W$ put

$$I_s = \{qr : sqr \in A_{\text{sp}}\}.$$

The canonical factorization is unique, so

$$|A_{\text{sp}}| = \sum_{s < W} |I_s|.$$

Lemma 4.2 (Pairwise overlap). *For $s \neq t < W$,*

$$|I_s \cap I_t| \ll \frac{n}{L\sqrt{Z} \log Z} + \frac{\sqrt{n}}{L}.$$

Proof. Use two labeled prime classes and orient every edge $qr \in I_s \cap I_t$ by $q > r$. The resulting simple bipartite graph is C_4 -free. A cycle with edges $q_i r_j, i, j \in \{1, 2\}$, would put the eight integers $sq_i r_j, tq_i r_j$ in A , and the four complementary pairs would all represent $stq_1 q_2 r_1 r_2$. Since $s, t < W < Z < q_i, r_j$, all eight integers are distinct. Moreover, all four edges satisfy $q_i > r_j$, so the two labeled vertex classes are numerically disjoint. Thus these are four genuine distinct-factor representations, a contradiction.

Fix a dyadic scale P for the smaller prime. The two vertex classes have sizes $O(P/\log P)$ and $O(n/(PL))$. The C_4 -free estimate therefore gives

$$O\left(\frac{n}{L\sqrt{P}\log P} + \frac{P}{\log P}\right)$$

edges. Summing over dyadic $P \geq Z$ proves the lemma. $\square$

Bonferroni's inequality and Lemma 4.2 give

$$\begin{aligned} |A_{\text{sp}}| &\leq \left| \bigcup_{s < W} I_s \right| + O\left(W^2 \left(\frac{n}{L\sqrt{Z}\log Z} + \frac{\sqrt{n}}{L}\right)\right) \\ &= \left| \bigcup_{s < W} I_s \right| + o(n/L), \end{aligned}$$

because $W^2 = L^{1/4}$ and $Z = L$. The union is contained in the set of semiprimes $qr \leq n$ with $q > Y$ and $r > Z$. Those with both primes at most $\sqrt{n}$ contribute $O(n/L^2)$. Otherwise the larger prime Q exceeds $\sqrt{n}$; if $j = \lfloor n/Q \rfloor$, the other prime lies in $(Z, j]$. Hence

$$|A_{\text{sp}}| \leq \sum_{j \geq 1} N_j(n)(\pi(j) - \pi(\min(j, Z))) + o(n/L). \quad (5)$$

4.2 The smooth form

Let A_{sm} be the elements of form (S), and define

$$T_q = \{t : qt \in A_{\text{sm}}\} \quad (q > \sqrt{n}).$$

For distinct primes q, Q ,

$$r_{T_q, T_Q}(m) \leq 2. \quad (6)$$

Indeed, three ordered pairs (u, v) with $uv = m$ would give three representations $(qu)(Qv) = qQm$. These representations have unequal factors and are distinct because $q, Q > \sqrt{n} > R$.

An *internal collision* is a pair of distinct unordered 2-multisets

$$\{a, b\} \neq \{c, d\}, \quad ab = cd, \quad a, b, c, d \leq R.$$

Diagonals such as $\{a, a\}$ are allowed. A fixed collision occurs in at most one fiber. If it occurred in both T_q and T_Q , two off-diagonal factorizations would give four ordered cross pairs, while a diagonal together with an off-diagonal factorization would give three. Both contradict (6).

There are at most R^4 possible collisions. For each collision that occurs, delete one tagged element (q, x) , meaning the copy of a support value x in the unique fiber T_q containing that collision, and call the cleaned fibers S_q . Then

$$\sum_q |T_q \setminus S_q| \leq R^4 = o(n/L), \quad (7)$$

and the whole family (S_q) , including each individual fiber, is compatible.

For each j with $N_j(n) > 0$, choose S_j of maximum size among the cleaned fibers in the bucket $\{q : \lfloor n/q \rfloor = j\}$; if $N_j(n) = 0$, put $S_j = \emptyset$. Choices from distinct nonempty buckets come from distinct prime fibers, so compatibility is preserved, and

$$|A_{\text{sm}}| \leq \sum_j N_j(n) |S_j| + o(n/L).$$

Finally set

$$U_j = S_j \cup \{p : Z < p \leq j, p \text{ prime}\}.$$

The two parts are disjoint. The family (U_j) is compatible because a product contains zero, one, or two prime factors exceeding Z ; these cases cannot collide, and compatibility or unique factorization gives at most two orientations within each case. Thus

$$|U_j| = |S_j| + \pi(j) - \pi(\min(j, Z)).$$

Restoring the $o(n/L)$ elements deleted in Lemma 4.1, and combining this identity with (5) and (7), gives

$$|A| \leq \sum_j N_j(n) |U_j| + o(n/L) \leq G(n) + o(n/L).$$

Together with Lemma 2.2, this proves Theorem 2.3.

5 The limit of the cofactor model

For a compatible family $\mathcal{U} = (U_j)$, put

$$w_j = \frac{1}{j(j+1)}, \quad e_j(\mathcal{U}) = |U_j| - 1 - \pi(j).$$

Definition 5.1. Define

$$\Gamma = \sup_{\mathcal{U} \text{ compatible}} \sum_{j \geq 1} w_j e_j(\mathcal{U}),$$

where a sum is assigned the value $-\infty$ if its negative part diverges.

Compatibility with $i = j$ makes each U_j a multiplicative Sidon set with diagonals allowed. The classical bound of Erdős [3], also recorded in [4, p. 252], is

$$|U_j| \leq \pi(j) + O(j^{3/4})$$

and therefore gives

$$e_j^+ \ll j^{3/4}, \quad \sum_{j \geq 1} w_j e_j^+ < \infty, \tag{8}$$

uniformly over compatible families. The family consisting of 1 and all primes has $e_j = 0$, so $0 \leq \Gamma < \infty$.

Proposition 5.2 (An explicit upper bound). *One has*

$$\Gamma \leq \zeta\left(\frac{7}{6}\right) + \frac{3}{2}\zeta\left(\frac{4}{3}\right) < 13.$$

Consequently

$$1 + M + \Gamma < 15.$$

Proof. We use the standard factor-graph argument, as in [8, Section 3.2]. Let $U \subseteq [N]$ be multiplicative Sidon.

First consider those $a \in U$ having a prime divisor $p > N^{2/3}$. Such a prime is unique, and we write $a = pb$, where $b < N^{1/3}$. Form the bipartite graph whose edges are these pairs (p, b) . Distinct edges represent distinct integers, and the graph is C_4 -free: a four-cycle would give four distinct elements satisfying

$$(pb)(qc) = (pc)(qb).$$

If d_p is the degree of p , then

$$\sum_p \binom{d_p}{2} \leq \binom{\lfloor N^{1/3} \rfloor}{2}.$$

Since $d_p - 1 \leq \binom{d_p}{2}$, this part of U has at most

$$\pi(N) - \pi(N^{2/3}) + \frac{1}{2}N^{2/3}$$

elements.

Every remaining $a \leq N$ has a factorization

$$a = uv, \quad v \leq u \leq N^{2/3}.$$

Indeed, this is immediate when $a \leq N^{2/3}$. Otherwise, if a has a prime divisor $N^{1/3} \leq p \leq N^{2/3}$, take the factors p and a/p . If all prime divisors are smaller than $N^{1/3}$, multiply prime factors until the product first exceeds $N^{1/3}$; that product and its complementary factor are both at most $N^{2/3}$. After interchanging the factors if necessary, also $v \leq \sqrt{a} \leq N^{1/2}$.

Fix one such factorization for each remaining element and form the corresponding bipartite graph. Its two vertex classes have sizes at most $\lfloor N^{2/3} \rfloor$ and $\lfloor N^{1/2} \rfloor$. Distinct edges represent distinct integers, so a four-cycle would again contradict the multiplicative Sidon property. The standard C_4 -free estimate [7] gives

$$e \leq N^{1/2} \sqrt{N^{2/3}} + N^{2/3} = N^{5/6} + N^{2/3}.$$

Combining the two parts gives

$$|U| - 1 - \pi(N) \leq N^{5/6} + \frac{3}{2}N^{2/3}.$$

Applying this to every fiber U_j , and using $w_j < j^{-2}$, gives

$$\Gamma \leq \zeta\left(\frac{7}{6}\right) + \frac{3}{2}\zeta\left(\frac{4}{3}\right).$$

For $s > 1$,

$$\zeta(s) < 1 + \frac{1}{s-1},$$

so

$$\Gamma < 7 + \frac{3}{2} \cdot 4 = 13.$$

Finally, the explicit cutoff estimate in the accompanying Lean development gives $M < 933/1000 < 1$, and therefore

$$1 + M + \Gamma < 15.$$

□

Lemma 5.3 (Canonical extension). *Every compatible prefix $U_1, \dots, U_J$ extends compatibly by*

$$U_j = U_J \cup \{p : J < p \leq j, p \text{ prime}\} \quad (j > J).$$

For this extension, $e_j = e_J$ for $j > J$.

Proof. Old elements are at most J , while every new prime exceeds J . For two fibers, products with no new prime are controlled by r_{U_i, U_J} or r_{U_J, U_J} , hence by the original prefix. With one new prime, unique factorization determines that prime and the old factor, leaving at most two orientations; with two new primes it again leaves at most two. Finally,

$$|U_j| = |U_J| + \pi(j) - \pi(J),$$

so $e_j = e_J$. □

Since $\sum_{j>J} w_j = 1/(J+1)$, the extension lemma implies

$$\sum_{j \leq J} w_j e_j \leq \Gamma + \frac{1 + \pi(J)}{J+1} = \Gamma + O(1/\log J). \quad (9)$$

Set

$$B(n) = \sum_{j \geq 1} N_j(n)(1 + \pi(j)).$$

Proposition 5.4. *As $n \rightarrow \infty$,*

$$G(n) = B(n) + (\Gamma + o(1)) \frac{n}{\log n}.$$

Proof. For fixed j , the prime number theorem gives

$$\frac{\log n}{n} N_j(n) \longrightarrow \frac{1}{j} - \frac{1}{j+1} = w_j. \quad (10)$$

The positive tails are uniform. In a dyadic block $K \leq j < 2K$, (8) and Chebyshev's bound give

$$\sum_{K \leq j < 2K} N_j(n) e_j^+ \ll K^{3/4} \pi(n/K) \ll \frac{n}{\log n} K^{-1/4}.$$

Summing the blocks yields

$$\frac{\log n}{n} \sum_{j > J} N_j(n) e_j^+ \ll J^{-1/4}. \quad (11)$$

Apply this to a family attaining $G(n)$. For fixed J , (10) is uniform over its prefix because $|e_j| \leq 2j+1$. Hence (9) and (11) give

$$\limsup_{n \rightarrow \infty} \frac{\log n}{n} (G(n) - B(n)) \leq \Gamma + O(1/\log J) + O(J^{-1/4}).$$

Let $J \rightarrow \infty$.

Conversely, choose a compatible family whose defining sum exceeds $\Gamma - \varepsilon$. Its defining series is absolutely convergent: the positive part is finite by (8), and the negative part is finite by Definition 5.1. Moreover, the trivial lower bound $e_J \geq -1 - \pi(J)$,

$$\frac{|e_J|}{J+1} \ll J^{-1/4} + \frac{1}{\log J}.$$

Choose J so that

$$\left| \sum_{j>J} w_j e_j \right| < \varepsilon \quad \text{and} \quad \frac{|e_J|}{J+1} < \varepsilon.$$

The canonical extension of this prefix then has weighted value

$$\sum_{j \leq J} w_j e_j + \frac{e_J}{J+1} > \Gamma - 3\varepsilon.$$

For this fixed extension,

$$\sum_{j>J} N_j(n) = \pi\left(\frac{n}{J+1}\right) - \pi(\sqrt{n})$$

once $n > (J+1)^2$. Together with (10), the prime number theorem gives

$$\frac{\log n}{n} \sum_j N_j(n) e_j \longrightarrow \sum_j w_j e_j.$$

Thus the lower limit is at least $\Gamma - 3\varepsilon$. Letting $\varepsilon \downarrow 0$ proves the proposition. $\square$

It remains to evaluate $B(n)$. Let $\pi_2(n)$ count products $pq \leq n$ with $p \leq q$ prime, including squares. Direct counting gives

$$\sum_j N_j(n) = \pi(n) - \pi(\sqrt{n}), \tag{12}$$

$$\sum_j N_j(n) \pi(j) = \pi_2(n) - \frac{\pi(\sqrt{n})(\pi(\sqrt{n}) + 1)}{2}. \tag{13}$$

The semiprime asymptotic [2, Theorem 1, equation (17)] is

$$\pi_2(n) = \frac{n \log \log n}{\log n} + (M + o(1)) \frac{n}{\log n}.$$

Consequently,

$$B(n) = \frac{n \log \log n}{\log n} + (1 + M + o(1)) \frac{n}{\log n}. \tag{14}$$

Theorem 2.3, Proposition 5.4, and (14) prove the asymptotic formula in Theorem 1.1. The bound $\Gamma \geq 4/15$ is proved next.

6 An explicit lower bound

Define the following seven fibers:

$$\begin{aligned} A &= \{1\}, & B &= \{1, 2\}, & C &= \{1, 2, 3\}, \\ D &= \{1, 3, 4, 5\}, & E &= \{1, 3, 4, 5, 6\}, & F &= \{1, 3, 4, 5, 6, 7\}, \\ & & & & G &= \{2, 3, 5, 7, 8, 9, 10\}. \end{aligned}$$

Assign the fibers

$$A, B, C, C, D, E, F, F, F, G$$

to $j = 1, \dots, 10$. The complete multiplication certificate in [Appendix A](#) proves that this prefix is compatible. Its excess sequence is

$$(e_1, \dots, e_{10}) = (0, 0, 0, 0, 0, 1, 1, 1, 1, 2).$$

The weighted prefix value is

$$\frac{1}{42} + \frac{1}{56} + \frac{1}{72} + \frac{1}{90} + \frac{1}{55} = \frac{14}{165}.$$

The canonical extension from G has $e_j = 2$ for $j > 10$, and contributes $2/11$. Therefore

$$\Gamma \geq \frac{14}{165} + \frac{2}{11} = \frac{4}{15}. \tag{15}$$

This completes the proof of [Theorem 1.1](#).

A The multiplication certificate

For two fibers X, Y , let

$$D(X, Y) = \{m : r_{X,Y}(m) = 2\}.$$

For each unordered pair of fiber types, the following table lists all products occurring more than once. Every listed product has convolution coefficient exactly two, and direct multiplication shows that every remaining product occurs at most once. Thus all 28 fiber-pair convolutions satisfy the compatibility condition.

Pair	Products with convolution coefficient exactly 2
AA, AB, AC, AD, AE, AF, AG	—
BB	2
BC	2
BD	—
BE	6
BF	6
BG	10
CC	2, 3, 6
CD	3
CE, CF	3, 6, 12
CG	6, 9, 10
DD, DE, DF	3, 4, 5, 12, 15, 20
DG	8, 9, 10, 15, 40
EE, EF	3, 4, 5, 6, 12, 15, 18, 20, 24, 30
EG	8, 9, 10, 12, 15, 30, 40
FF	3, 4, 5, 6, 7, 12, 15, 18, 20, 21, 24, 28, 30, 35, 42
FG	8, 9, 10, 12, 15, 21, 30, 35, 40
GG	6, 10, 14, 15, 16, 18, 20, 21, 24, 27, 30, 35, 40, 45, 50, 56, 63, 70, 72, 80, 90

References

- [1] Thomas F. Bloom, *Erdős problem 796*, Erdős Problems, <https://www.erdosproblems.com/796>, accessed 14 July 2026.
- [2] Dragoş Crişan and Radek Erban, *On the counting function of semiprimes*, Integers **21** (2021), Paper No. A122, 13.
- [3] Paul Erdős, *On sequences of integers no one of which divides the product of two others and on some related problems*, Mitt. Forsch.-Inst. Math. Mech. Univ. Tomsk **2** (1938), 74–82.
- [4] Paul Erdős, *On the multiplicative representation of integers*, Israel Journal of Mathematics **2** (1964), no. 4, 251–261.
- [5] Paul Erdős, *Problems and results in chromatic graph theory*, Proof Techniques in Graph Theory (Frank Harary, ed.), Academic Press, New York, 1969, Proceedings of the Second Ann Arbor Graph Theory Conference, Ann Arbor, Michigan, 1968, pp. 27–35. MR 0252273
- [6] Leo Goldmakher, *A quick proof of Mertens' theorem*, Online note, Available at <https://web.williams.edu/Mathematics/lg5/mertens.pdf>, accessed 15 July 2026.
- [7] Tamás Kővári, Vera T. Sós, and Pál Turán, *On a problem of K. Zarankiewicz*, Colloquium Mathematicum **3** (1954), no. 1, 50–57.
- [8] Hong Liu and Péter Pál Pach, *The number of multiplicative Sidon sets of integers*, Journal of Combinatorial Theory, Series A **165** (2019), 152–175.
- [9] Quanyu Tang, *On Erdős Problem 796*, Online note, January 2026, Available at [GitHub repository](#), accessed 14 July 2026.